

Network Packet Sniffer with Alert System Report

1. Introduction

Network traffic monitoring is an important part of cybersecurity and network management. Attackers often perform activities such as port scanning, packet flooding, and suspicious traffic generation before launching an attack. A packet sniffer is a tool that captures and analyzes network packets in real time. This project focuses on building a Network Packet Sniffer with an Alert System to monitor network traffic and detect anomalies such as port scanning and flooding attacks.

2. Abstract

This project presents a Network Packet Sniffer developed using Python and Scapy to capture and analyze network packets in real time. The system logs packet information such as IP address, port number, packet length, and flags. The captured data is stored in an SQLite database for analysis. The system detects anomalies such as port scanning and traffic flooding based on predefined thresholds. When suspicious activity is detected, the system generates alerts through logs or email notifications. This project helps in understanding network traffic analysis and intrusion detection techniques.

3. Tools Used

Python, Scapy, SQLite Database, Matplotlib, Email/Alert System, Tkinter (optional GUI).

4. Steps Involved in Building the Project

1. Install Python and required libraries such as Scapy, SQLite, and Matplotlib. 2. Capture network packets using Scapy. 3. Extract packet header information such as IP address, port, packet size, and flags. 4. Store captured packet data in an SQLite database. 5. Analyze traffic patterns to detect anomalies such as port scanning and packet flooding. 6. Set threshold values for suspicious activity detection. 7. Generate alerts when thresholds are exceeded. 8. Display traffic summary using Matplotlib graphs. 9. Optionally, create a GUI for live traffic monitoring and alerts.

5. Conclusion

The Network Packet Sniffer with Alert System successfully captures and analyzes network traffic and detects suspicious activities such as port scanning and flooding. The system stores packet data in a database and generates alerts when abnormal traffic is detected. This project improves understanding of packet analysis, intrusion detection, and network monitoring systems. The project can be further enhanced by adding machine learning-based anomaly detection and a real-time dashboard.